

FARGO HECTOR, ND, USA

WMO: 727530

Lat: 46.925N

Lon: 96.811W

Elev: 274

StdP: 98.07

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.2	-25.5	-31.8	0.2	-27.6	-29.2	0.3	-25.2	15.1	-14.6	13.3	-11.8	3.3	310	0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.2	22.4	30.6	21.3	29.1	20.5	24.2	29.7	23.0	28.5	21.9	27.3	6.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.7	28.0	21.1	16.3	26.8	20.0	15.2	25.5	74.6	29.8	69.8	28.5	65.6	27.4	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.6	11.3	10.2	DB	-31.4	35.0	3.5	1.5	-33.9	36.1	-36.0	37.0	-37.9	37.8	-40.5	38.9	(4)
(5)			WB	-31.3	26.3	3.0	1.1	-33.4	27.0	-35.2	27.7	-36.9	28.3	-39.0	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.0	-12.6	-10.3	-2.6	6.3	13.7	19.4	21.9	20.6	16.1	8.0	-0.9	-8.9	(6)
(7)		DBStd	13.47	7.98	7.61	7.48	5.85	5.10	3.81	3.20	3.28	4.82	5.27	6.40	7.13	(7)
(8)		HDD10.0	2844	701	568	392	137	22	0	0	0	7	102	330	584	(8)
(9)		HDD18.3	4825	959	802	648	361	158	31	8	15	99	324	578	843	(9)
(10)		CDD10.0	1379	0	0	3	27	138	284	368	328	188	39	2	0	(10)
(11)		CDD18.3	317	0	0	0	1	16	64	118	85	31	3	0	0	(11)
(12)		CDH23.3	3150	0	0	1	17	204	629	1146	811	311	31	0	0	(12)
(13)		CDH26.7	1001	0	0	0	3	64	191	386	251	98	7	0	0	(13)
(14)	Wind	WSAvg	4.9	5.0	5.1	5.2	5.5	5.4	4.7	4.1	4.1	4.6	5.2	5.1	5.0	(14)
(15)	Precipitation	PrecAvg	544	17	14	29	44	70	90	71	63	58	47	21	18	(15)
(16)		PrecMax	883	47	44	117	134	200	298	196	191	165	179	116	47	(16)
(17)		PrecMin	225	2	2	3	0	12	15	11	5	3	1	0	2	(17)
(18)		PrecStd	134	13	10	19	31	44	58	45	38	37	40	22	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	8.0	19.1	25.7	32.0	33.7	34.1	33.6	32.6	27.3	17.3	7.4	(19)
(20)			MCWB	2.4	5.0	12.8	13.9	18.8	22.9	24.2	23.6	21.7	16.7	11.0	4.7	(20)
(21)		2%	DB	2.4	4.4	13.8	21.9	28.3	30.9	31.9	31.3	29.5	22.5	13.3	3.9	(21)
(22)			MCWB	0.8	2.5	9.3	12.1	17.1	20.9	23.2	22.0	20.1	14.5	8.5	1.5	(22)
(23)		5%	DB	0.7	2.5	10.2	18.9	25.3	28.8	30.4	29.5	27.0	19.0	10.3	1.9	(23)
(24)			MCWB	-0.6	1.0	6.9	10.6	15.7	19.5	22.2	21.1	18.7	12.8	6.7	0.4	(24)
(25)		10%	DB	-1.3	0.7	7.0	16.2	22.8	27.1	28.9	27.8	24.4	16.3	7.5	0.3	(25)
(26)			MCWB	-2.4	-0.3	4.5	9.4	14.2	18.7	21.2	20.2	17.6	11.1	4.5	-0.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	5.7	14.0	15.3	20.7	24.6	26.2	25.2	23.4	18.1	12.4	4.5	(27)
(28)			MCDWB	4.3	7.4	17.2	22.8	27.8	30.7	31.0	31.4	29.3	24.4	16.5	7.0	(28)
(29)		2%	WB	0.9	2.7	9.9	12.8	18.8	22.9	24.7	23.7	21.6	15.7	9.2	1.8	(29)
(30)			MCDWB	2.2	4.3	14.0	20.0	25.8	28.8	30.3	29.2	27.5	21.1	12.8	3.5	(30)
(31)		5%	WB	-0.5	1.1	6.9	11.3	17.0	21.3	23.4	22.4	20.1	13.4	6.7	0.5	(31)
(32)			MCDWB	0.7	2.4	9.9	17.8	22.8	26.9	28.7	27.5	24.9	18.3	10.2	1.8	(32)
(33)		10%	WB	-2.4	-0.4	4.4	9.8	15.4	20.0	22.3	21.2	18.6	11.5	4.7	-0.8	(33)
(34)			MCDWB	-1.4	0.9	6.9	15.5	21.1	24.9	27.6	26.1	23.6	15.6	7.4	0.3	(34)
(35)	Mean Daily Temperature Range	MDBR	9.0	9.4	9.2	11.2	12.3	11.3	11.6	12.1	12.1	10.2	8.8	8.2	(35)	
(36)		5% DB	MCDBR	9.6	9.3	13.3	16.5	15.9	13.5	13.1	13.8	14.9	14.7	12.4	9.8	(36)
(37)			MCWBR	8.4	7.8	9.3	9.0	8.1	6.4	6.4	6.5	7.2	8.4	8.5	8.2	(37)
(38)		5% WB	MCDWB	9.3	9.0	12.9	14.7	13.2	11.3	11.7	11.9	12.5	13.1	11.8	9.2	(38)
(39)			MCWBR	8.3	7.6	9.3	8.6	7.5	6.4	6.8	6.6	7.0	8.0	8.3	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.306	0.306	0.361	0.379	0.392	0.392	0.396	0.360	0.318	0.287	0.278	(40)	
(41)		taud	2.225	2.274	2.433	2.332	2.347	2.350	2.343	2.330	2.419	2.516	2.504	2.377	(41)	
(42)		Ebn at Noon	766	858	924	893	886	873	867	848	851	845	810	771	(42)	
(43)		Edh at Noon	82	96	96	117	120	121	120	117	97	76	62	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.66	2.81	3.87	4.90	5.50	6.11	6.42	5.38	4.02	2.51	1.55	1.17	(44)	
(45)		RadStd	0.17	0.30	0.40	0.41	0.37	0.41	0.26	0.44	0.27	0.36	0.17	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (27 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page